

Name	No Forced Pre-emption (server)	Voluntary Kernel Pre-emption (desktop)	Preemptible Kernel (low latency desktop)	Fully Preemptible Kernel (real-time)
Kernel compilation	Compiled kernel without applied RT-patch	Compiled kernel without applied RT-patch	Compiled kernel without applied RT-patch	Compiled kernel with applied RT-patch
Hackbench load	hackbench -l 100000000	hackbench -l 100000000	hackbench -l 100000000	hackbench -l 100000000
Cyclictest duration	10 min	10 min	10 min	10 min
Latency	13548us	541us	699us	156us

Table 2: Cyclictest observations

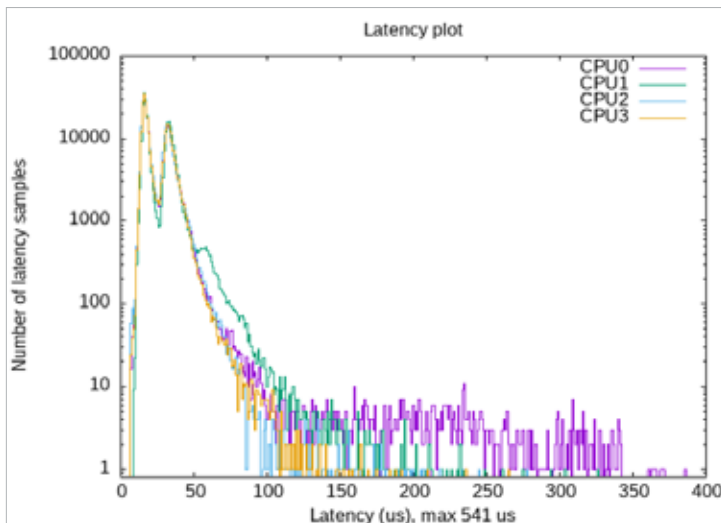


Figure 7: Graph of Voluntary Kernel Preemption

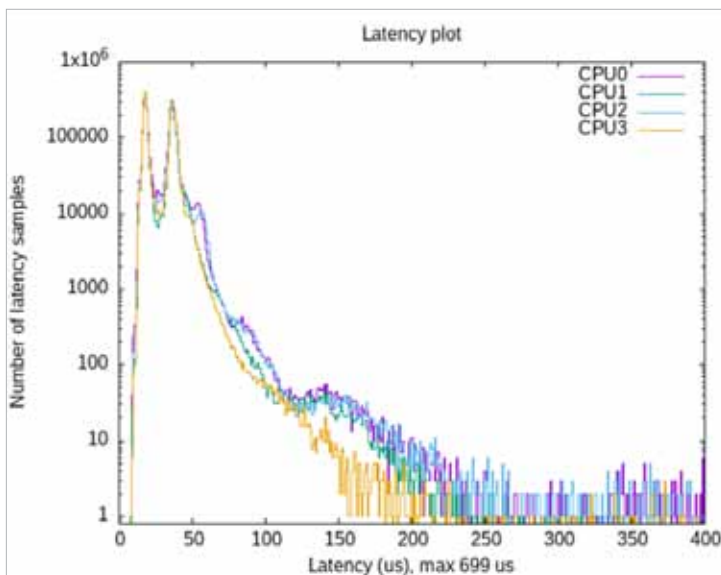


Figure 8: Graph of Preemptible Kernel

Application guidelines

- **Real-time thread priority:** Real-time threads should be assigned the highest priority (SCHED_FIFO or SCHED_RR) to ensure they are not preempted by lower-priority tasks.
- **Use POSIX real-time APIs:** Leverage the POSIX real-time APIs, such as pthreads with SCHED_FIFO, for managing real-time threads. These APIs provide mechanisms for setting thread priorities, managing synchronisation, and avoiding priority inversion.
- **Minimise blocking and I/O:** Real-time threads should minimise blocking operations, especially those that may involve waiting on external events. If blocking is necessary, use non-blocking I/O or asynchronous I/O mechanisms.
- **Avoid busy-waiting:** Avoid busy-waiting (e.g., spinlocks) as it can lead to high CPU usage and increased system latency. Instead, use blocking mechanisms or conditional waiting (e.g., pthread_cond_wait) to efficiently manage synchronisation.
- **Manage memory carefully:** Minimise dynamic memory allocation during real-time operation as it can introduce non-deterministic behaviour. Allocate memory statically or use memory pools to ensure deterministic access.
- **Resource management:** Properly manage resources, such as file descriptors, network connections, and hardware peripherals, to avoid contention and resource exhaustion. Release resources when they are no longer needed.